SECP3744 - UTHM CASE STUDY

UTHM INVENTORY SYSTEM PROJECT REPORT

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DECLARATION

I declare that this project report entitled *"INTEGRATED INVENTORY MANAGEMENT SYSTEM FOR UTHM: A CASE STUDY""* is the result of my own research except as cited in the references. The project report has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

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ABSTRACT

Enterprise Architecture (EA) plays a crucial role in aligning information systems with organisational strategies, particularly within higher education institutions that rely heavily on integrated digital platforms. Universiti Tun Hussein Onn Malaysia (UTHM) has adopted EA principles to support its academic, administrative, and research operations through various subsystems. However, challenges related to system integration, interoperability, and architectural consistency continue to affect operational efficiency and user experience. This study aims to analyse and enhance the existing enterprise architecture framework at UTHM to better support institutional objectives.

The System Development Life Cycle (SDLC) methodology was employed to guide the planning, analysis, design, and implementation phases of the proposed architectural improvements. Data were collected through document analysis, system observation, and stakeholder input to identify current limitations and requirements. Based on the findings, an improved enterprise architecture model was designed to address identified gaps, focusing on better system integration, scalability, and governance.

The outcome of this study demonstrates that the application of a structured SDLC approach can effectively support the enhancement of enterprise architecture in a higher education environment. The proposed improvements contribute to improved system alignment, reduced redundancy, and more efficient service delivery for students and staff. This study provides valuable insights for higher education institutions seeking to strengthen their digital transformation initiatives through systematic enterprise architecture development.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Universiti Tun Hussein Onn Malaysia (UTHM) is in the process of constructing a new on-premise data centre to facilitate the consolidation, modernisation, and oversight of its institutional IT infrastructure. Notwithstanding this strategic commitment, the institution does not own a formalised Enterprise Architecture (EA) that can deliver a cohesive framework aligning business processes, information systems, data assets, and foundational technical platforms. The lack of a formalised Enterprise Architecture has resulted in disjointed system implementations, segregated data repositories, restricted visibility of inter-system relationships, and considerable difficulties in achieving interoperability and effective data integration across many departments and functional areas. These conditions impair operational efficiency, restrict data-driven decision-making, and obstruct UTHM’s capacity to attain sustainable digital transformation. In acknowledgement of these limitations, UTHM has demonstrated a willingness to accept external and academic contributions that can present a complete and adaptable EA model appropriate for institutional application.

This thesis addresses the institutional requirement by developing a comprehensive Enterprise Architecture for UTHM, informed by The Open Group Architecture Framework (TOGAF). TOGAF is utilised for its globally acknowledged structured methodology, focus on governance, and appropriateness for large, complex entities like higher education institutions. The proposed EA aims to create a cohesive and standardised architectural framework that aligns UTHM's strategic goals with its technological capacities, enhances integration among academic and administrative systems, improves governance and lifecycle management of IT assets, and facilitates scalable future digital initiatives. The study further creates a prototype enterprise system utilising SAP technology to operationalise and test the suggested design. This prototype illustrates the actual application of essential architectural elements, highlights the integration of enterprise resource processes within the EA framework, and assesses the viability of SAP as a fundamental platform to underpin UTHM's mission-critical operations.

This research seeks to furnish UTHM with a systematic, pragmatic, and future-oriented Enterprise Architecture model by integrating architectural design and system prototyping, thereby reinforcing institutional governance, fostering system interoperability, improving data integration, and advancing the university's enduring digital modernisation strategy within its new data centre environment.

1.2 Problem Statement

UTHM is in the process of establishing a new on-premise data center to support the centralization and modernization of its institutional IT infrastructure. UTHM currently does not possess a functional EA that clearly represents how its business processes, applications, data, and technology components are integrated and governed. This is because UTHM lacks EA expertise.

The lack of a well-defined EA has resulted in fragmented system management, limited visibility of system interdependencies, and difficulties in ensuring effective data integration. In order to solve this problem, UTHM has stated that it is open to external proposals, including academic contributions, that could demonstrate a suitable enterprise architecture model for potential adoption.

In response to this challenge, this case study focuses on improving the current system landscape in order to design an enhanced and functional Enterprise Architecture for UTHM. In addition, a prototype system utilizing SAP technologies is developed to allow UTHM to test an integrated enterprise system within the proposed architecture. This approach aims to demonstrate how SAP can be aligned with the EA to support core university operations, improve data integration, and validate the practicality of the proposed architecture within UTHM's new data center environment.

1.3 Project Objectives

- To design a comprehensive architectural framework that aligns the inventory system with UTHM’s organizational structure and ICT policies.
- To model and automate business processes that allow seamless handovers between departments.
- To establish a Single Source of Truth by designing an integrated enterprise data model and eliminating data redundancy and ensuring consistent asset information across all UTHM divisions.
- To develop a system capable of generating enterprise-level insights, helping UTHM management make data-driven decisions on budget planning and asset utilization.

1.4 Project Scope

The scope of this project is designed to fit in the project where this project focuses on providing a structured transition from UTHM's existing enterprise model to an integrated enterprise model based on their goals in digitalizing transformation. The main components of the project are:

1. Review existing system enterprise model

This project begins with an "As-Is" assessment of UTHM's current system environment based on the visit institutional observations. This phase focuses on analyzing how redundant processes and the lack of real-time data sharing hinder operational efficiency. The goal is to document the current relationship between business processes and evaluate how the current system lack of integration impacts institutional operational efficiency.

2. Design To-Be Enterprise Model

The project involves designing a comprehensive "To-Be" Enterprise Model that provides a scalable and integrated blueprint for the university that maps the four architectural layers (Business, Data, Application, and Technology) The scope includes a detailed design of Use Case Diagrams, Use Case Descriptions, Activity Diagrams, and Sequence Diagrams to model the behavioral logic of the subsystems. Additionally, it covers Database Design through Class Diagrams and Entity Relationship Diagrams (ERD) to ensure data integrity, and designing the interface to define the user experience and interaction for the individual subsystem within the new architecture.

3. System Development

This project also involves building a working prototype using the SAP Business Technology Platform to connect different departments and show how information flows smoothly. In this stage, the physical database is created to store all necessary data, and the main system functions are coded. This process demonstrates how an automated, integrated system makes university operations faster and more efficient.

1.5 Project Importance

This project is important as it helps UTHM address the current problems especially in UTHM's enterprise architecture at the same time to improve its operational efficiency, system integration and user experience. The project ensures that data flows smoothly across platforms like Learning Management System (LMS) and student portal by analyzing and enhancing the integration of subsystems such as student enrollment, course registration, hostel management and university healthcare. This improvement surely benefits students, staff and administrators by providing them with reliable access to information and services.

Optimising proposal enterprise architecture helps UTHM make better decisions, use resources more effectively, and ensure their IT systems can grow over time. This approach will take full advantage of new technologies like AI, IoT, and cloud computing to support their academic, administrative, and research work. In the end, this project strengthens UTHM's commitment to digital transformation, operational excellence, and ongoing improvement, helping them remain a modern and efficient university.

CHAPTER 2

LITERATURE REVIEW

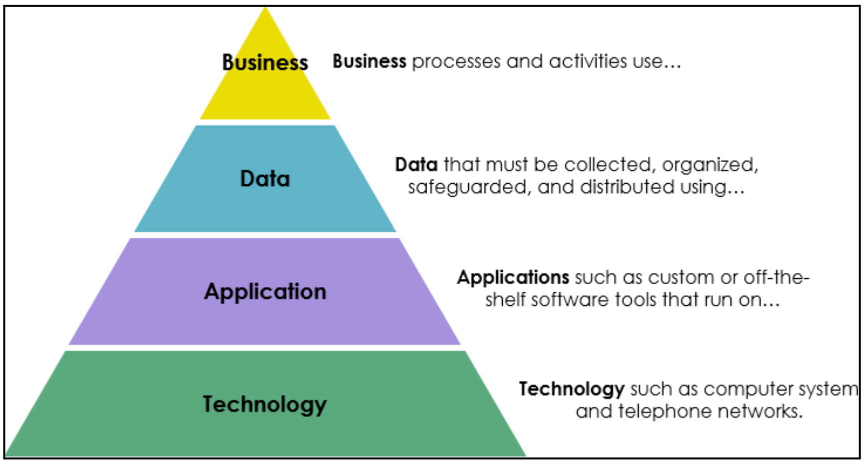
2.1 Introduction

This chapter provides a literature review of the key concepts that define the enterprise-level design and integration discussed in this project. The modern enterprise, especially in higher education, requires a comprehensive framework to strategically align its business objectives with its Information Technology (IT) capabilities and infrastructure. This framework is known as Enterprise Architecture (EA). We will also discuss the related sub-system that exist in universities EA by taking UiTM as an example. In the end, we are going to discuss technology used in EA which is the SAP Business Technology Platform and its details.

2.2 Enterprise Architecture (EA)

Enterprise Architecture (EA) is defined as a comprehensive framework designed to manage and align an organization's business objectives with its Information Technology (IT) capabilities and infrastructure (BOC Group, 2023; CIO Index, n.d.). It encompasses the systematic design, planning, and management of the organization's technology, processes, and systems to effectively support its strategic goals and objectives (BOC Group, 2023). EA provides a holistic view of the enterprise by organizing its structure into four interconnected architectural domains (Visual Paradigm, 2024)

Figure 1: *The Four Domains of Enterprise Architecture*



Note. Source: (Visual Paradigm, 2024).

Business Architecture: This domain maps the organization's strategy, governance, core business capabilities, value streams, and processes. Its central focus is on defining the "what" and "why" of the business operations, ensuring that technological investments are driven by strategic needs (CIO Index, n.d.).

Data Architecture: This domain establishes the principles and structures for how the organization's data assets are to be collected, stored, managed, and utilized. It is critical for ensuring data quality, consistency, and accessibility throughout the entire enterprise (BOC Group, 2023).

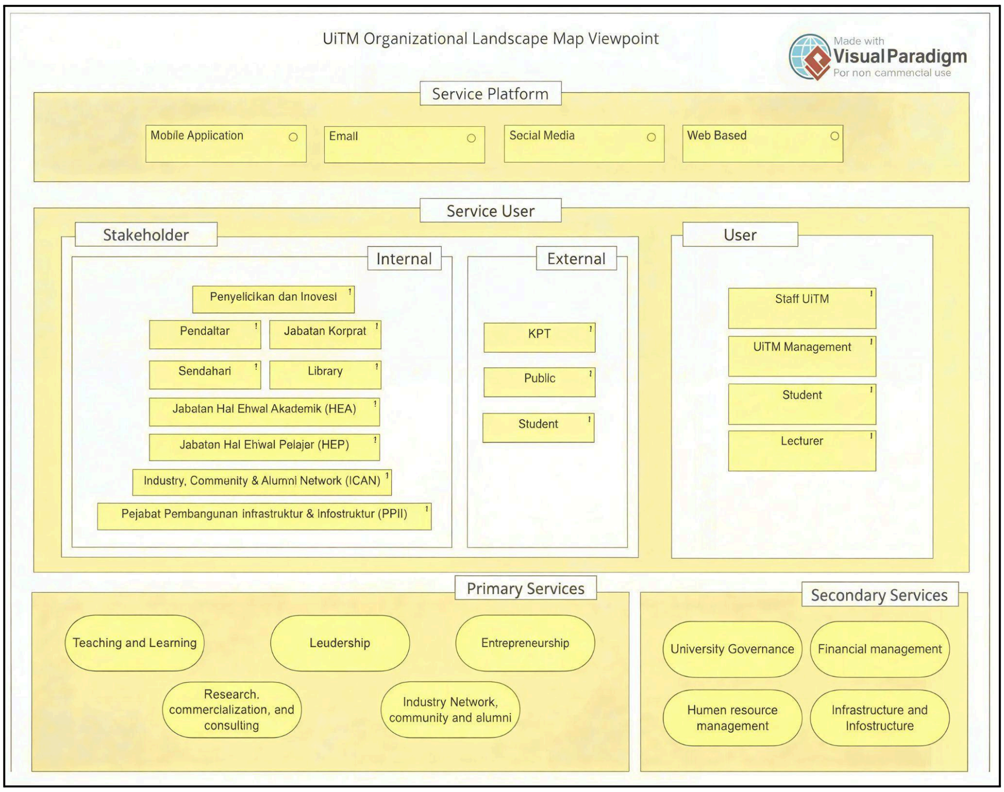
Application Architecture: This domain involves identifying the specific applications utilized within the enterprise and defining their interactions with each other and with external stakeholders. The objective is to establish an integrated, efficient, and scalable application portfolio (CIO Index, n.d.).

Technology Architecture: This domain describes the foundational hardware, software, networking infrastructure, and operating systems that collectively support the application and data layers. Its primary concern is ensuring the reliability, scalability, and security of the underlying technology foundation (BOC Group, 2023; CIO Index, n.d.).

2.3 Related Subsystem

Enterprise Architecture (EA) in higher education institutions typically integrates multiple subsystems that support academic, research, governance, administration, and community functions. These subsystems interact across different layers, including business processes, service delivery, information technology platforms, and stakeholder interactions, in an attempt to align institutions with their strategic goals. Many universities adopt EA for simplifying services provided to students and staff, standardising information management, and increasing decision-making effectiveness (Amin et al., 2024).

Figure 2: UiTM Organizational Landscape Map Viewpoint



Note: Source: (Amin et al., 2024).

Figure 2.2 shows UiTM's Enterprise Architecture (EA) landscape, which is built around different subsystems. Each subsystem handles specific tasks, such as academic services, administrative services and digital platforms. They all work together to connect stakeholders and service users. The framework includes internal and external stakeholders, main and supporting service areas and works across platforms like mobile, email, social media, and websites. By linking these subsystems, UiTM creates a smooth and integrated service environment that supports its goals for improving the institution (Amin et al., 2024).

Subsystem Category	Subsystem	Description
Service Platforms	Mobile App, Email, Social Media, Web-based Systems	Digital channels enabling communication and service access for users.
Stakeholders	Internal: HEA, HEP, Library, Corporate, ICAN, PPII, etc.	Organizational units involved in academic, student, infrastructure, alumni, and corporate services.
	External: KPT, Public, External Students	External bodies influencing or consuming UiTM services.
User Groups	UiTM Staff, Management, Students, Lecturers	Primary service users interacting with UiTM digital and administrative services.
Primary Services	Teaching & Learning, Leadership, Entrepreneurship, Research & Commercialization, Alumni & Industry Network	Core mission-driven services supporting UiTM's academic and developmental goals.
Secondary Services	University Governance, Human Resource, Finance, Infrastructure	Supporting subsystems enabling operations, compliance, and resource management.

Table 2.1 Description of UiTM EA Subsystems

Table 2.1 provides an overview of how each subsystem at UiTM supports the services offered at the university and the involvement of different stakeholders. It categorises subsystems into functional types, describes their purpose, and identifies service platforms on which users can access them. It also provides information on the internal and external stakeholders and user groups, and the primary and secondary services each subsystem enables. This structured view shows how UiTM integrates its digital and administrative systems into a seamless, mission-driven service environment.

Feature	Layered EA Model	UiTM Organizational Landscape Map Viewpoint
Main Focus	**Architectural Components** (What are the components of the enterprise's IT?).	**Organizational and Service Components** (Who uses the services and what services are offered?).
Business/Organizational Content	Lists **Functional Areas**: Academic Management, Finance, Human Resources, etc.	Lists **Key Stakeholders** (Internal/External), **Users** (Staff, Student, Lecturer), and **Services** (Teaching, Leadership, Governance).
IT/Technology Content	Lists **Abstract Domains**: Servers, Network, Middleware, Database (under Technology Architecture).	Lists **Delivery Channels/Platforms**: Mobile Application, Email, Social Media, Web Based (under Service Platform).
Framework Link	Strongly aligned with the **TOGAF/Zachman** frameworks' core layered structure.	Represents a **Viewpoint** or business-centric model, often part of an overall EA deliverable.

Table 2.2 Comparison between Layered EA Model and UiTM Organizational Landscape Map Viewpoint

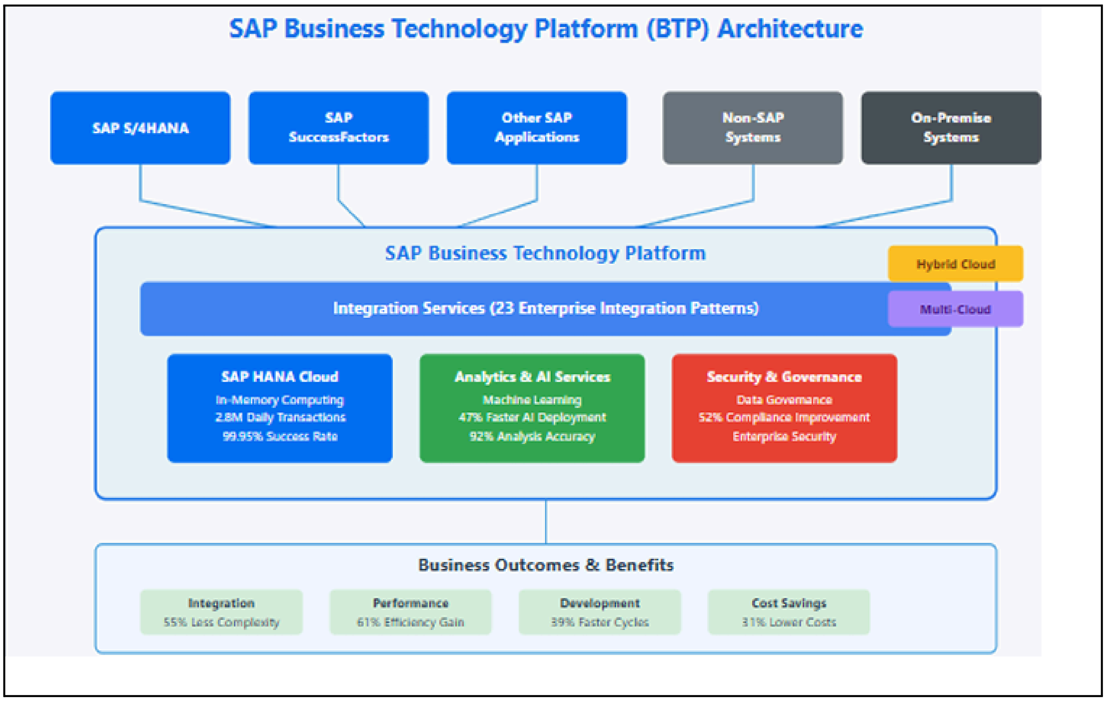
This Table 2.2 visual two representations of Enterprise Architecture (EA) serve distinct but complementary roles: the Layered EA Model acts as a structural blueprint, primarily focusing on Architectural Components by detailing the required Functional Areas like Academic Management and the underlying abstract IT Domains such as Servers and Network, consistent with the formal, tiered structure of frameworks like TOGAF; conversely, the UiTM Organizational Landscape Map Viewpoint functions as a high-level, business-centric communication tool, focusing on Organizational and Service Components by illustrating the key Stakeholders

(Staff, Students) and the Delivery Channels (Mobile Apps, Web) they use to access the university's services.

2.4 Technology Used

The SAP Business Technology Platform (BTP) is architecturally designed as a unified cloud foundation, acting as a crucial integration layer for the modern enterprise (Komarina & Sajja, 2025). It is built upon a framework that encompasses the Integration Suite, Extension Suite, and Data Management Suite, leveraging a microservices approach to provide a platform with defined scalability features and a structured security framework (Annanki, 2023).

Figure 4 : *SAP BTP Architecture*



Note: Source: (Komarina & Saja, 2025).

Core Components of SAP BTP

1. **Database and Data Management**
   Database and data management focuses on efficient data storage, processing, and governance. Key services include SAP HANA Cloud, SAP Data Warehouse Cloud, and SAP Data Intelligence, which collectively support real-time data access and enterprise-level data management.

2. **Analytics**
   SAP BTP includes analytical tools such as SAP Analytics Cloud (SAC), enabling organizations to perform reporting, data visualization, predictive analytics, and planning. These tools help businesses make informed, data-driven decisions.

3. **Application Development and Automation**
   This component offers capabilities for building and extending applications using low-code/no-code tools, cloud-native development frameworks, and process automation. Tools such as SAP Build Apps, SAP Build Process Automation, and the SAP Cloud Application Programming Model (CAP) streamline application development and workflow automation.

4. **Integration**
   SAP BTP supports seamless connectivity between SAP and non-SAP systems through services like SAP Integration Suite, API Management, and Event Mesh. These tools ensure smooth data exchange and interoperability across the organization's technology landscape.

SAP BTP enables organizations to modernize business processes, reduce dependency on custom code in core systems, and create flexible, scalable solutions. By unifying application development, data management, analytics, and integration services, the platform helps businesses enhance innovation, improve efficiency, and adapt quickly to changing operational needs.

2.5 Summary

This chapter establishes the theoretical foundation for the project by exploring Enterprise Architecture (EA) as a tool to align university business goals with IT infrastructure. By examining the four key domains which are Business, Data, Application, and Technology, the review clarifies how a system must be structured to support an entire organization.

Through the case study of UiTM's EA landscape, the chapter illustrates how various subsystems and stakeholders interact to create an integrated service environment. Finally, the review of SAP Business Technology Platform (BTP) identifies the modern technology needed to achieve this integration, focusing on data management, analytics, and seamless system connectivity. Together, these elements provide the blueprint for designing the UTHM Inventory System as a robust enterprise solution.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

This chapter is a description of the methodology of Waterfall with System Development Life Cycle (SDLC) used to help with the planning, analysis, design and implementation of improvements to UTHM's Enterprise Architecture framework. The Waterfall of SDLC uses formal, creating or altering systems, and the models and methodologies that people use to develop these systems (Kute & Thorat, 2014). The choice to use a structured methodology focused on requirements, architectures and specifications being fully defined and validated earlier in the process was due to the increased difficulty in changing or correcting Enterprise Architecture projects after they have been implemented due to their high cost of implementation and complexity to implement from start to finish. The following chapters explain each phase of the methodology, including phases, activities and deliverables, provide a project schedule like for example using a Gantt Chart or Kanban format with major milestones, and summarise the entire development of the project.

3.2 Proposed Methodology

This study used the Waterfall System Development Life Cycle (SDLC) model as its approach. The Waterfall methodology is chosen for its systematic, documentation-centric, and linear characteristics, making it appropriate for major business projects like the creation of business Architecture (EA) for Universiti Tun Hussein Onn Malaysia (UTHM). Considering that EA projects entail substantial implementation costs, widespread institutional effects, and long-term strategic coherence, a methodology prioritising meticulous planning, requirement verification, architectural delineation, and regulated execution is necessary. The Waterfall SDLC facilitates thorough analysis and robust architectural design prior to deployment, reducing rework and maintaining architectural consistency throughout the project lifecycle.

This study uses the Waterfall technique to direct the design and development of an improved Enterprise Architecture framework for UTHM, alongside the creation of an enterprise-wide SAP prototype to validate the proposed architecture. Each phase of the technique concentrates on systematically converting UTHM's institutional requirements into a structured design, subsequently validated through technical implementation. Stakeholder interaction is integrated to guarantee conformity with institutional expectations and operational reality. The suggested methodology has five primary phases: Feasibility Study, System Analysis, System Design, Implementation, and Maintenance and Support.

3.3 Phases of Methodology

The project follows the five core phases of the SDLC which is Feasibility Study, System Analysis, System Design, Implementation and Maintenance and support

Phase 1:Feasibility Study

This first stage will determine whether the project is viable. We adhere to UTHM's standard operating procedure for new system requests.

- FSR Process: The project began with the submission of the Feasibility Study Report (FSR).
- Committee Approval: FSR is submitted to the IT Security & Steering Committee which will evaluate the proposal.

Outcome: The "Go /No-Go" decision based on the project proposal and literature review.

Phase 2: System Analysis

In this phase , we collect the detailed business requirements to bridge the gap between the definition of the requirement and how it should be.

- Data Collection: Conducting interviews with UTHM stakeholders (Asset Unit officers) to understand the current manual form and workflow.
- Requirement Specification: Defining the user roles and their access privileges.

Phase 3: System Design

In this phase , we convert the requirements into technical specifications.

- Architecture Design: We utilize the three-tier architecture model to ensure the system is scalable and easy to maintain.
- Database Design:Create the entity-relationship diagram to model the relationships between asset, staff and location.
- Interface Design: Prototyping the dashboard UI to ensure they are user-friendly for non technical staff.
- Infrastructure Design: The system will be hosted in UTHM's Data Center. It will benefit from high-availability infrastructure that includes dual power sources, UPS backup and generator set to ensure very high uptime.

Phase 4: Implementation

This is the phase where the system is built and implemented.

- Coding: Developing the subsystem using web technologies such as PHP and java then connect it to the database MySql.
- Data Migration: Adopt a phased migration strategy, which converts sample data from existing Excel sheets into the new system format.
- Security Implementation: The code will be hosted behind the UTHM's existing security layer which include 2 Firewalls (Europe/China) and a Web Application Firewall to protect the portal from being attacked from the external side.
- Testing: Perform testing on each individual module and integration testing to ensure it communicates with the Financial module.

Phase 5: Maintenance and Support

In this phase , we involve training and patching activities. We will teach the UTHM staff how to use the new QR code features and fix any bugs that are discovered after the system is deployed.

3.4 Project Planning Schedule

In order to make sure this project runs smoothly, a schedule consists of every phase and its duration is made. The team members that are in charge of leading the task are also determined to make sure all the phases are supervised. Both Figure 3.1 and Table 3.1 show the Gantt Chart and task with the assigned members.

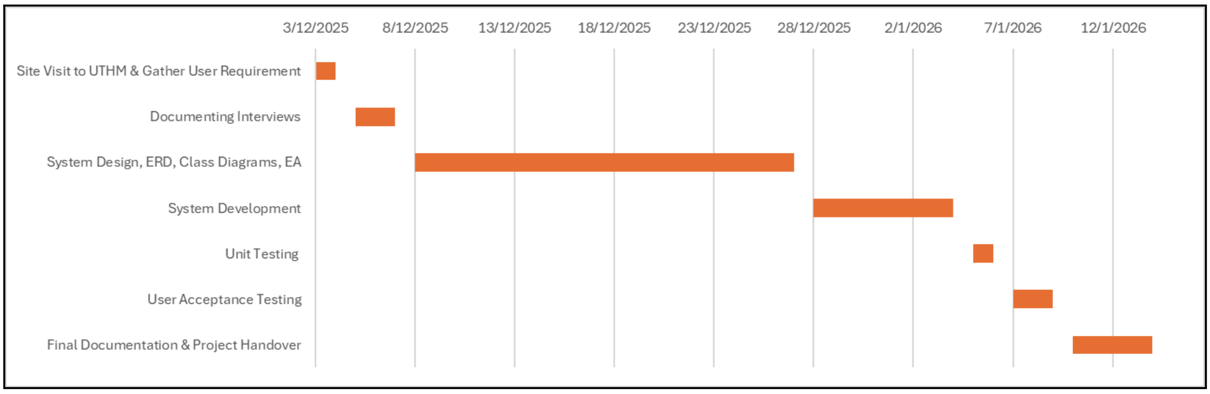


Figure 3.1 Project Schedule Gantt Chart

Tasks	Start Date	End Date	Task Leader
Site Visit to UTHM & Gather User Requirement	3/12/2025	4/12/2025	Aisyah
Documenting Interviews	5/12/2025	7/12/2025	Harini
System Design, ERD, Class Diagrams, EA	8/12/2025	27/12/2025	Zhiming
System Development	28/12/2025	4/1/2026	Afiq
Unit Testing	5/1/2026	6/1/2026	Auni
User Acceptance Testing	7/1/2026	9/1/2026	Ziyaad
Final Documentation & Project Handover	10/1/2026	14/1/2026	Ika

Table 3.1 Task Distribution Assign with Task Leader

3.5 Summary

This chapter has detailed the Waterfall System Development Life Cycle (SDLC) methodology chosen to guide the improvement of UTHM’s Enterprise Architecture. By utilizing the five core phases which are feasibility study, system analysis, system design, implementation, and maintenance, the project minimizes the risks associated with the high cost and complexity of enterprise-level systems.

This structured path ensures that the resulting SAP prototype is not only technically sound but also strategically aligned with UTHM's specific operational needs and secure within its new data center environment. The inclusion of a detailed project schedule and Gantt chart ensures that all tasks, from the initial site visit to system implementation and testing are supervised and completed within the established timeframe. Ultimately, this structured methodology serves as the bridge between UTHM’s strategic digital transformation goals and the successful delivery of a secure, integrated, and high-performance enterprise environment.

CHAPTER 4

ANALYSIS AND DESIGN

4.1 Introduction

This chapter analyses the organisational structure of Universiti Tun Hussein Onn Malaysia (UTHM), the current system environment, and enterprise requirements to facilitate the development of a complete Enterprise Architecture (EA) framework and a SAP-based enterprise prototype. A comprehensive comprehension of the university's organisational structure, operational processes, and information dissemination methods is crucial for ensuring that the suggested design is congruent with institutional goals and effectively tackles existing operational issues. Given that UTHM presently functions with several autonomous systems exhibiting minimal integration, this analysis establishes the essential groundwork for delineating how a cohesive and centralised enterprise framework might improve institutional efficiency, governance, and data-informed decision-making.

The chapter initially analyses UTHM's organisational framework at the Strategic, Tactical, and Operational tiers to pinpoint principal decision-makers, stakeholders, and primary data generators. This is succeeded by a comprehensive evaluation of the existing system landscape, emphasising the dependence on disjointed digital platforms and semi-manual procedures, although the presence of a contemporary three-tier data centre architecture. A comparative analysis of the current and proposed environments is subsequently provided to demonstrate the shift from isolated systems to a fully integrated enterprise ecosystem, facilitated by centralised data management, Single Sign-On (SSO), and an improved user experience.

To guarantee that the suggested architecture appropriately represents institutional demands, requirements gathering activities were performed through stakeholder involvement via structured interviews. These interactions facilitated the recognition of essential business requirements, operational obstacles, technical anticipations, and governance factors. The chapter finishes by delineating the functional and non-functional requirements that will inform the design, development, and deployment of the improved EA framework. These investigations collectively establish the analytical foundation for the ensuing architectural design and solution implementation phases.

4.2 Company Organization Structure

A successful system integration requires clear understanding of the organizational hierarchy and the information flow between stakeholders. UTHM operates on three distinct levels which are Strategic, Tactical and Operational. This structure must be bridged to eliminate the "Vertical Silos" which are often found in large organizations.

1. Strategic Level
    - IT Security and Steering Committee: This high-level body responsible for the overall digital strategy for UTHM. They will review and approve the Feasibility Study Reports for all proposed systems. They need to determine whether the new system is necessary and aligns with the university budget and culture.
    - Management Implications: Top management support is critical to enforce the cultural changes to adapt the new system.
2. Tactical Level
    - Bursary Department (Bendahari) : Bursary defines the rules that the software must obey. They act like a central hub for data verification, requiring inputs from various faculty to ensure accuracy.
3. Operational Level
    - Faculties and Responsibility Centers: This level includes academic and administrative staff that are distributed across the campus. They are the primary data generators.

4.3 Current System Analysis

UTHM relies on a range of digital systems and technologies to support its academic, administrative, and operational activities. Online platforms are used for teaching and learning, student administration, finance, human resource management, and communication between students and staff. These systems enable basic information access and daily operations. However, they are implemented independently to meet specific departmental needs. As a result, data sharing across departments are limited and operational efficiency depends on manual processes. In terms of infrastructure, UTHM is in the process of developing a new three-tier data center to host and support its digital systems. While this infrastructure provides a strong physical foundation for digital services, its capabilities have not yet been fully utilized in the current system environment as there are no centralized system management and integrated digital solutions. This situation highlights the need for improved system coordination and structured planning to maximize the benefits of the new data center.

4.4 Comparison between Existing and Proposed System

The transition from UTHM's current digital landscape to the proposed architecture represents a shift from a fragmented system to a unified digital system. Currently, the university relies on independent systems that serve specific departmental needs. While a modern three-tier data center exists, its potential remains not fully utilized because the current systems lack the centralized management and integration needed to utilize such a foundation. The proposed system addresses these limitations by replacing manual workarounds with end-to-end digitalization, guided by a clear "Digital Transformation" vision. Instead of isolated data, the proposed architecture introduces a centralized data warehouse and automated pipelines to ensure information flows seamlessly across the institution. This change moves towards a more agile, hybrid cloud environment. Furthermore, the user experience is improved from using multiple disconnected platforms to a streamlined approach featuring Single Sign-On (SSO) and a unified mobile app.

4.5 System Requirements Gathering Technique

To ensure the proposed UTHM Inventory System aligns with the actual operational needs of the university, the team employed the Interview Method as the primary data collection technique. This qualitative approach allowed the team to explore the complexities of existing workflows and identify specific pain points across different hierarchical levels. The interview method was selected due to its flexibility and the depth of information it provides. Compared to other methods like questionnaires, interviews allow the team to capture the specific needs of both the administrative staff and the end-users. The interviews were conducted during the site visit on 3/12/2025. The team roles were distributed as follows:

- **Lead Interviewer (Aisyah):** Facilitated the discussion using a prepared set of open-ended questions.
- **Technical Observers (Afiq & Ziyaad):** Focused on identifying technical constraints and integration points with current UTHM legacy systems.
- **Note-Takers (Ika & Harini):** Documented the responses and captured non-verbal cues regarding system frustrations.

Below are the sample of questions interviewed and its answers:

Q: Are there still manual methods or all is digitalized?

A: UTHM is not 100% digital yet. One of the departments that are still running manually are their Health Center. However, the admin has been enforcing the lecturer to use LMS and work on a singular system.

Q: Is there a stand alone database different from the main database?

A: Only LMS has separated database but all the system get data in on single database

Q: Who approves of the new system?

A: The team asks for user requirements first, then the team will develop the system and send it to the committee to decide either to accept or reject the new system.

4.6 System Requirements

Functional Requirements

Functional requirements specify the particular functionality and features that an updated enterprise architecture (EA) framework should offer to UTHM to help streamline primary business operations in UTHM’s IT ecosystem. Functional requirements help to ensure that the updated EA framework provides support for key UTHM processes such as asset management, decision making and compliance.

- User Authentication - A secure account accessible for all UTHM staff and students using Single Sign-On (SSO).
- Requirements Management - All systems requirements should be captured, traced and validated from the SDLC analysis phase and then linked to the design output.
- Reporting - Generate customizable reports that should be exportable along with Gantt charts and phase deliverables that can be viewed by administrators.
- Integration Support - EA Framework must provide the ability to import and export data that is in existing tools like TOGAF during integration activities.

Non-Functional Requirements

Non-functional requirements specify the quality attributes and performance standards the EA framework must meet, ensuring reliability, efficiency, and usability in UTHM's demanding academic environment.

- Performance: System shall load web pages within 3 seconds for datasets up to 1,000 components.
- Scalability: System must support around 2000 concurrent users.
- Security: Encrypt data especially for passwords and using firewalls like Web Application Firewall (WAF) on the webpage system.
- Usability: Intuitive for new users and accessible for various users.
- Reliability: The system must have 99.5% uptime.

4.7 System Design

4.7.1 Enterprise Architecture AS-IS

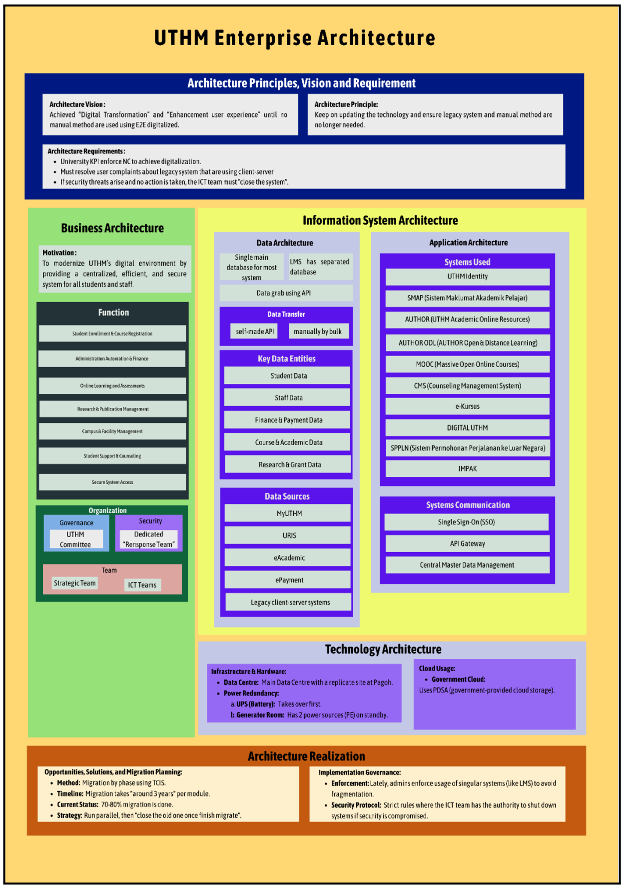


Figure 4.1 UTHM Enterprise Architecture As-Is

Figure 4.1 is UTHM EA As-Is that combines all the answers from the questionnaire and staff from UTHM.

4.7.2 System Design AS-IS

This section explains the current architecture at UTHM before the new system is implemented.

1. Business Architecture

- **Strategy:** Guided by "Digital Transformation 2030" to shift the culture from manual to digital, but administrative tasks still require manual forms currently.
- **Governance**: All proposed systems must pass the IT Steering Committee via a Feasibility Study Report and approved based on necessity and department expertise.
- **Operations**: Departments operate in horizontal silos. The LMS and TCIS are currently run as separate business entities with disconnected workflow.

2. **Application Architecture**
   - **Core system**: Legacy TCIS based on client-server technology.
   - **Accessibility**: Restricted on campus LAN. Staff cannot access the system while outstation without a VPN , this leads to bad user experience.
   - **Integration**: Applications are fragmented. For example the LMS is a separate application from the main student database, requiring API to bridge the gap.

3. **Data Architecture**
   - **Storage**: Data is distributed and decentralized. For example the student marks and physical asset record are separate, siloed databases.
   - **Synchronization**: The data transfer relies on API calls or need manual entry , rather than real-time synchronization.
   - **Performance**: Current database structure struggles with high concurrency or usage, causing up to 5- minute lags during peak usage.

4. **Technology Architecture**
   - **Facility**: Tier 3 Data Center located on the upper floor (flood prevention).
   - **Server Hosting**: Uses Hot Aisle/Cold Aisle Containment for thermal efficiency.
   - **Power**: High availability via Dual Power Sources (TN A/TN B), backed by UPS and Generator Sets.

- **Security**: Perimeter protection using 2 Firewalls (Europe/China brands) and a Web Application Firewall (WAF) for portal defense.

4.7.3 Enterprise Architecture To-Be

Figure 4.2 illustrates the Enterprise Architecture To-Be for UTHM. It represents the future system structure of the university. This architecture helps improve system efficiency and supports UTHM's future digital development.

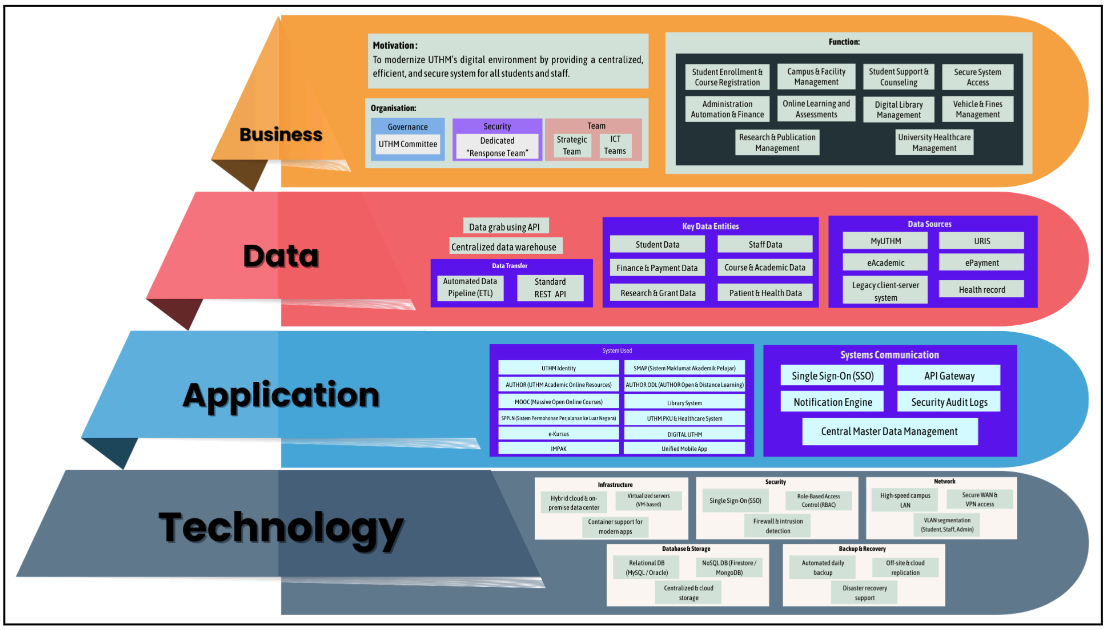


Figure 4.2 UTHM Enterprise Architecture To-Be

4.7.4 To-Be

This section explains the Enterprise Architecture To-Be as shown in Figure 4.2. The framework is organized into four main layers. Each layer serves a specific purpose.

1. Business Architecture

   This layer, shown in green, describes what the university does and how it operates:

   - Motivation: To develop a centralized, secure and efficient digital environment accessible to students and staff on any device at any time.
   - Organization: Illustrates UTHM's departmental hierarchy.
   - Functions: The architecture supports ten core business functions:

Business Functions	Description
Student Enrollment & Course Registration	Manages student admissions, course selection, and registration processes in a streamlined manner.
Administration Automation and Finances	Manages administrative tasks, payroll, payment processing and allocation of university resources.
Online Learning and Assessment	Support student learning and coursework using online platforms.
Research and Publication Management	Supports research project management, grant applications and innovation initiatives.
Campus and Facility Management	Coordinates maintenance, space allocation, and utilization of campus facilities.
Student Support and Counselling	Manages student welfare services, counselling sessions, and academic support programs.
Secure System Access	Authorized access to systems through identity management and authentication mechanisms.
University Healthcare Management	Supports the management of healthcare services and medical records within the university.
Digital Library Management	Provides access to digital academic resources, journals and library services.
Vehicle and Fines Management	Manages vehicle registration, parking systems and fine enforcement processes.

Table 4.1 Business Functions Description

2. Information System Architecture

   Divided into Data Architecture and Application Architecture:

   - Data Architecture: Includes a centralized data warehouse that consolidates data from multiple systems. It supports dashboards for data visualization, automated data pipeline and data protection mechanisms. REST APIs are used to enable secure and standardized data exchange between systems.

- Application Architecture: The Application Architecture consists of multiple applications that support academic, administrative, and operational activities at UTHM.

Activities	Application
Student & Academic Systems	UTHM Identity, SMAP (Sistem Maklumat Akademik Pelajar), AUTHOR (UTHM Academic Online Resources), AUTHOR ODL (Author Open and Distance Learning), MOOC (Massive Open Online Courses) and online academic resources.
Administrative & Support Systems	Student Counselling System, SPPLN, UTHM PKU and Healthcare System, e-Kursus, DIGITAL UTHM, IMPAK, and Vehicle and Facilities Management systems.
Library & Access Systems	Library System and Unified Mobile Application.
Integration	Connects all systems through Single Sign-On (SSO), API Gateway, notification engine, security audit logs, and centralized master data management

Table 4.2 Application Layer

3. Technology Architecture

   Defines the infrastructure supporting all systems and is organized into five domains:

Domains	Description
Infrastructure	Hybrid cloud access and on-premise data center with virtualization support to ensure scalability, flexibility and efficient resource utilization.
Network	High-speed backbone network, secure Wan and VPN access to support secure and advanced connectivity

	across the campus.
Database	Relational (MySQL/Oracle), NoSQL (Firebase/MongoDB) and cloud storage for high availability.
Security	Single Sign-On (SSO) , secure internal communications, firewalls, and intrusion detection to protect applications, data, and network resources.
Backup	Automated backups, offsite/cloud replication, and disaster-recovery support data protection and business continuity.

Table 4.3 Domains in Technology Architecture

4.7.5 System Architecture

Figure 4.3 below illustrates the cloud system architecture of the UTHM Inventory Subsystem. The diagram shows users accessing the system through laptops and mobile devices, while the database is hosted in the cloud using SAP to enable centralized data storage and access.

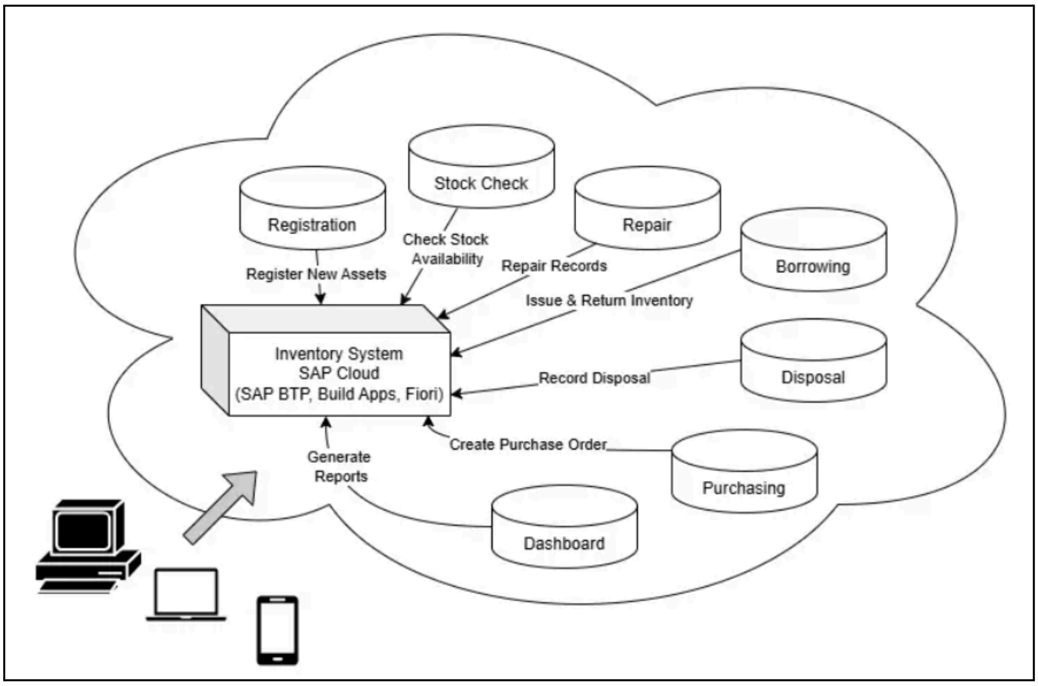


Figure 4.3 UTHM Inventory Cloud System Architecture

4.7.6 System Architecture Components

This section describes the system architecture of the UTHM Inventory System. As illustrated in Figure 4.3, the system is built on a cloud-based inventory database hosted on SAP Cloud. It centralizes all operations to enable real-time access and efficient asset management.

The SAP Cloud serves as the main hub for inventory management and implemented in several SAP tools, including SAP Build Apps, SAP Business Technology Platform (BTP) and SAP Fiori. Through this hub, the cloud manages all key inventory functions, such as:

1. **Registration:** Adding new assets to the system.
2. **Stock Check:** Monitoring the availability of items.
3. **Repair:** Recording maintenance and repair activities.
4. **Borrowing:** Managing the issuance and return of items.
5. **Disposal:** Logging assets that are no longer in use.
6. **Purchasing:** Creating orders and tracking procurement processes.
7. **Dashboard & Reporting:** Generating comprehensive reports and visual summaries for decision-making.

Users can access the system through computers, laptops, and mobile devices. This flexible multi-device access supports remote inventory management, increasing convenience and operational efficiency.

4.7.7 Project Design

In this individual project design, disposal items are selected from the existing inventory system once they are identified as damaged or no longer usable. The selected items go through a disposal request, approval, and recording process to ensure proper control, accountability, and documentation.

Use Case Diagram

Figure 4.4 illustrates the use case diagram of the Disposal module within the Inventory System. It shows how different actors interact with the system to manage the disposal of inventory items. The main actors involved are Inventory Staff, Supervisor, and Disposal Committee.

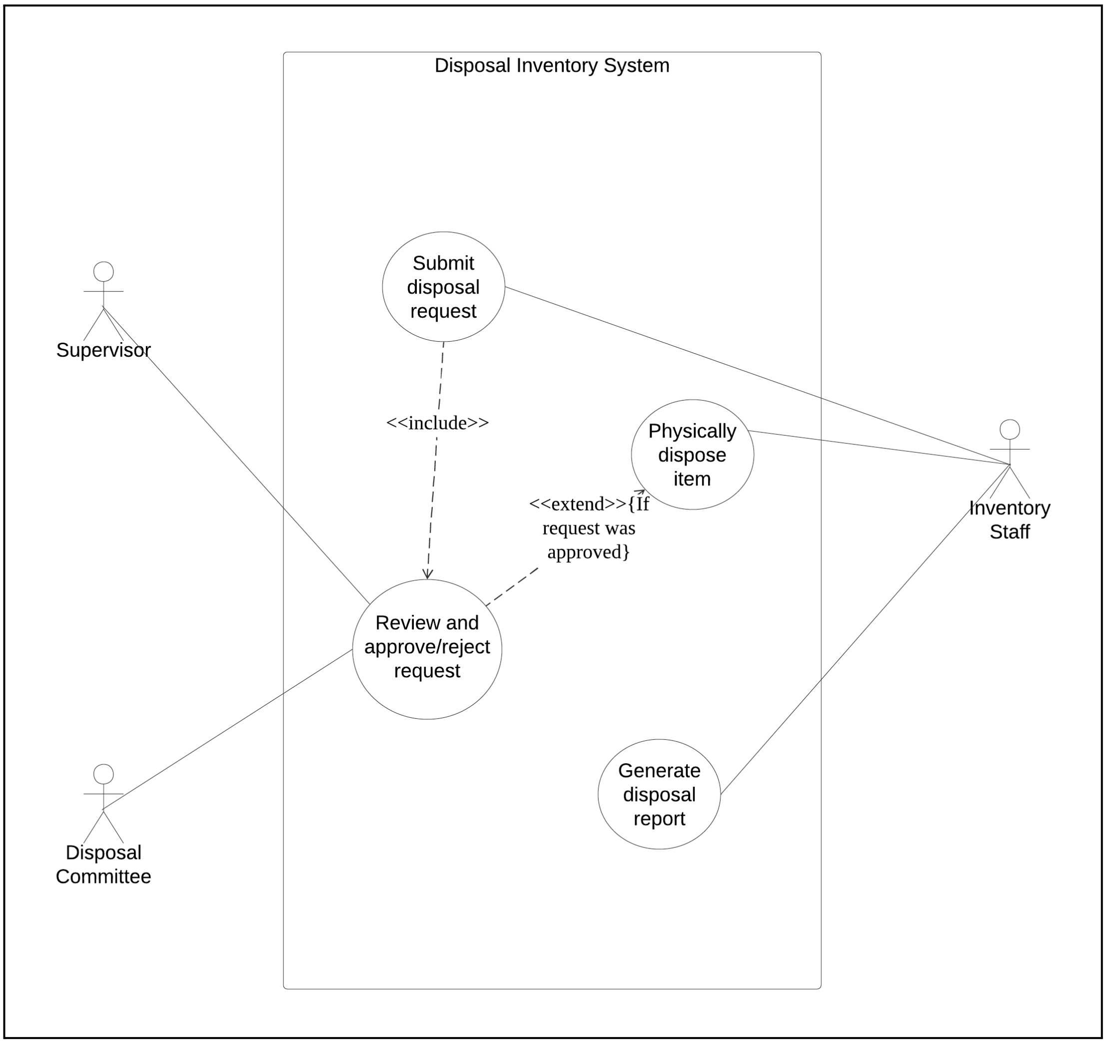


Figure 4.4 Use Case Diagram of Disposal Inventory System

Use Case Description

A use case description explains how a user or actor interacts with a system to achieve a specific goal, detailing step-by-step interactions, success paths or main flow, alternative paths, and failure paths or exceptions in a narrative format to define system requirements from the user's perspective.

Use Case: Submit Disposal Request
ID: UC-D1
Actor: Inventory Staff
Preconditions: 1. The inventory system already been logged in using SSO 2. The item exists in the inventory database and has status “Damaged” which is eligible for disposal.
Flow of events: 1. The use case starts when Inventory Staff decides that a damaged item should be disposed of. 2. Inventory Staff opens the “Disposal Request” page. 3. Inventory Staff select the item from the inventory list by item ID. 4. Inventory Staff fills in disposal details or reasons. 5. Inventory Staff submit the disposal request. 6. The system forwards the request to the Supervisor/Disposal Committee for review.
Postcondition: 1. A disposal request is stored in the system with status “Pending Approval”.

Table 4.4 Use Case Description of Submit Disposal Request

Table 4.4 presents a structured use case description with the id UC-D1 for the Submit Disposal Request process in an inventory management system, where Inventory Staff initiates disposal of damaged items. Preconditions require the item to already be logged in the system and marked as "Damaged" in the inventory database. The main flow begins with the staff deciding on disposal, opening the Disposal Request page, filling in item details and reasons by ID, and submitting the request, which the system then forwards to the Supervisor/Disposal Committee for review while storing it with "Pending Approval" status.

Use Case: Review and Approve or Reject Request
ID: UC-D2
Actor:

Supervisor and Disposal Committee
Preconditions: 1. The inventory system already been logged in using SSO 2. The item exists in the inventory database and there is at least one disposal request with status “Pending Approval”.
Flow of events: 1. The use case starts when the Supervisor decides to review pending disposal requests. 2. The system displays a list of pending requests. 3. Supervisor/Disposal Committee selects a specific request to review. 4. The system shows requested item details, damage information, and justification for disposal. 5. Supervisor/Disposal Committee decides to approve or reject the request. 6. The system updates the request status to “Approved” or “Rejected”.
Postcondition: 1. The disposal request has a final decision status “Approved” or “Rejected” recorded in the system.

Table 4.5 Use Case Description of Review and Approve or Reject Request

Table 4.5 serves as a Use Case Description with an id of UC-D2, providing a structured flow for how a Supervisor or Disposal Committee interacts with an inventory system to process disposal requests. It defines the necessary preconditions, ensuring the user is authenticated via SSO and that pending requests exist, before detailing a six-step flow of events that moves from reviewing item justifications to making a final decision. The process is designed to ensure data integrity by ending with a postcondition where the system successfully updates and records the request status as either "Approved" or "Rejected".

Use Case: Physically Dispose Item
ID: UC-D3
Actor: Inventory Staff
Preconditions: 1. The inventory system already been logged in using SSO 2. The item exists in the inventory database and has a disposal request for the item that has status “Approved”.
Flow of events: 1. The use case starts when Inventory Staff needs to carry out the approved disposal. 2. Inventory Staff reviews the approved disposal request and instructions.

3. Inventory Staff disposes the item according to the approved method and university policy. 4. Inventory Staff records disposal details in the system. 5. The system saves the disposal record.
Postcondition: 1. The item has been physically disposed of and the disposal action is recorded in the database of the system.

Table 4.6 Use Case Description of Physically Dispose Item

Table 4.6 describes the use case for physically disposing of an inventory item with the id UC-D3 in an inventory management system. The flow begins when the Inventory Staff initiates the disposal task, reviews the approved request and its instructions, and then physically disposes of the item according to the approved method and university policy. After completing the disposal, the staff records the disposal details in the system, which then saves the disposal record. The use case ends with the postcondition that the item has been successfully disposed of and the action is officially recorded in the inventory system's database.

Use Case: Generate Disposal Report
ID: UC-D4
Actor: Inventory Staff
Preconditions: 1. The inventory system already been logged in using SSO 2. The item exists in the inventory database and contains disposal records.
Flow of events: 1. The use case starts when Inventory Staff needs a report of disposal activities. 2. Inventory Staff opens the "Disposal Report" application in the system. 3. Inventory Staff confirms the selected filters and requests to generate the report. 4. The system retrieves matching disposal records from the database. 5. The system generates the disposal report in the selected format.
Postcondition: 1. A disposal report is generated and available to Inventory Staff for review, download, or printing.

Table 4.7 Use Case Description of Generate Disposal Report

Table 4.7 explains the use case for generating a disposal report with the. The process begins when the Inventory Staff needs a report of disposal activities and accesses the "Disposal Report" application in the system. The staff then selects and confirms the desired filters, after which the system retrieves the relevant disposal records from the database and generates the report in the selected format. The use case concludes with the report being successfully generated and made available for the Inventory Staff to review, download, or print.

Activity Diagram and Sequence Diagram

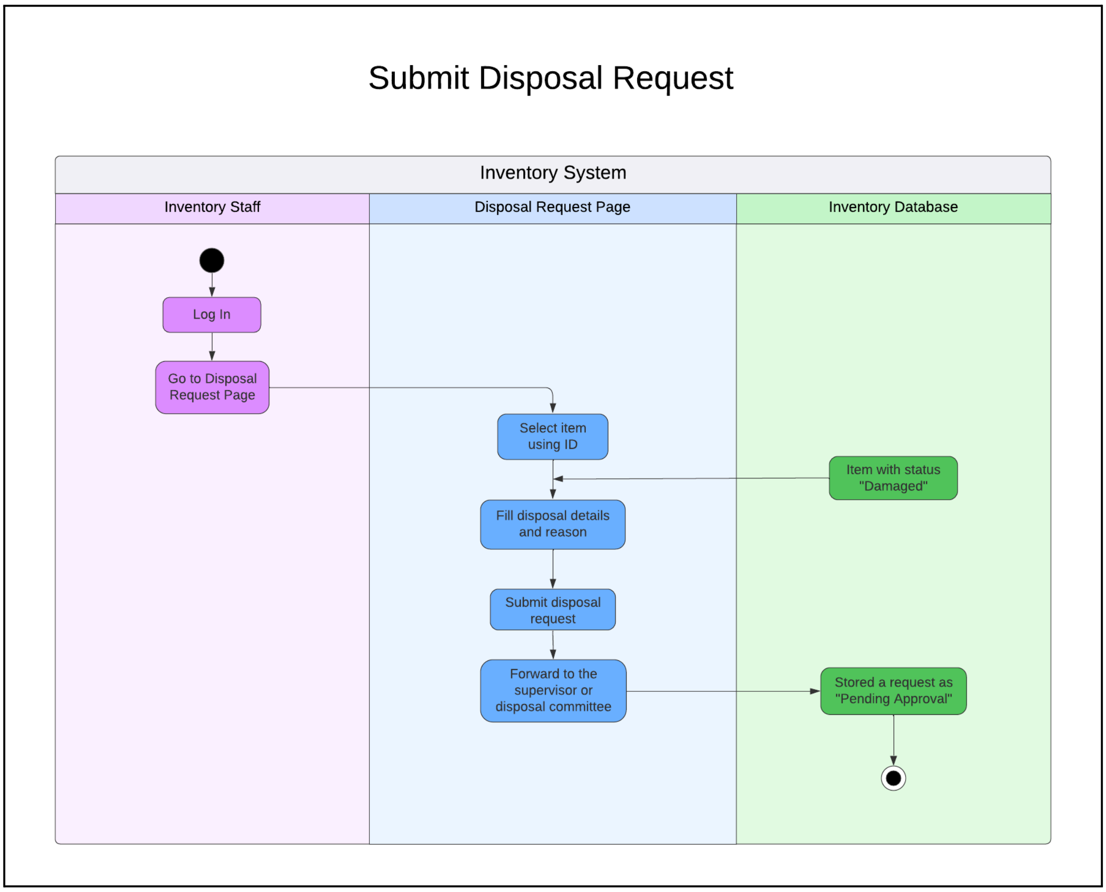


Figure 4.5 Activity Diagram of Submit Disposal Request

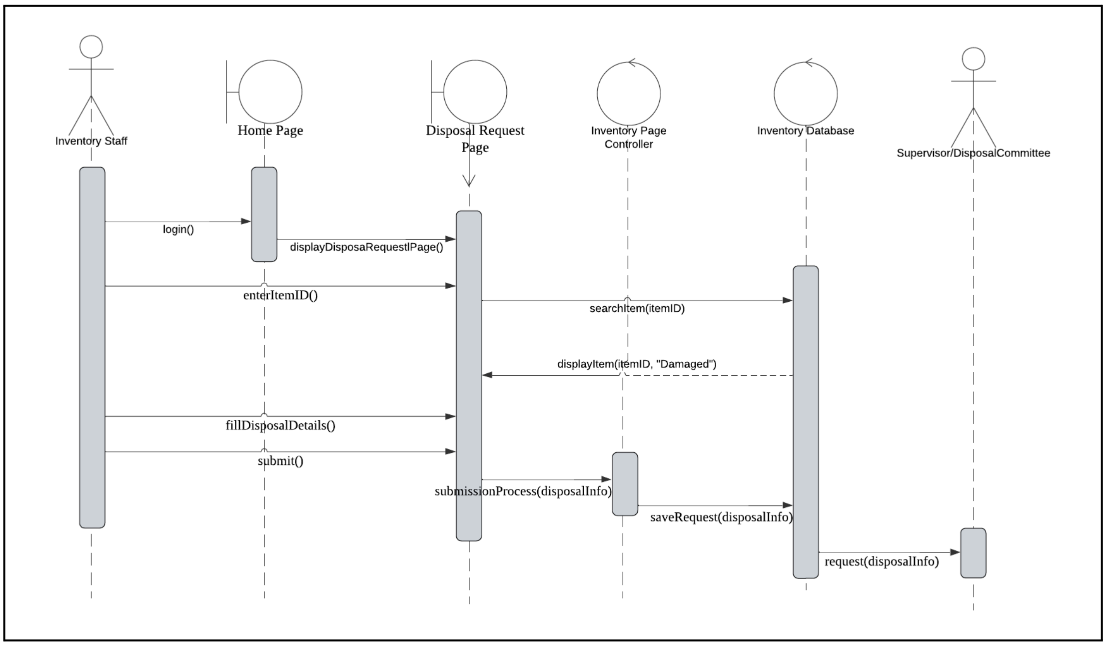


Figure 4.6 Sequence Diagram of Submit Disposal Request

Figure 4.5 and figure 4.6 above shows a workflow that starts with the Inventory Staff selecting a specific item on the Disposal Request Page then the system validates that the item is already marked as "Damaged" in the database. The staff then fills in the disposal details and reason for the request. Upon submission, the system forwards the request to a supervisor/disposal committee and saves it in the Inventory Database as "Pending Approval."

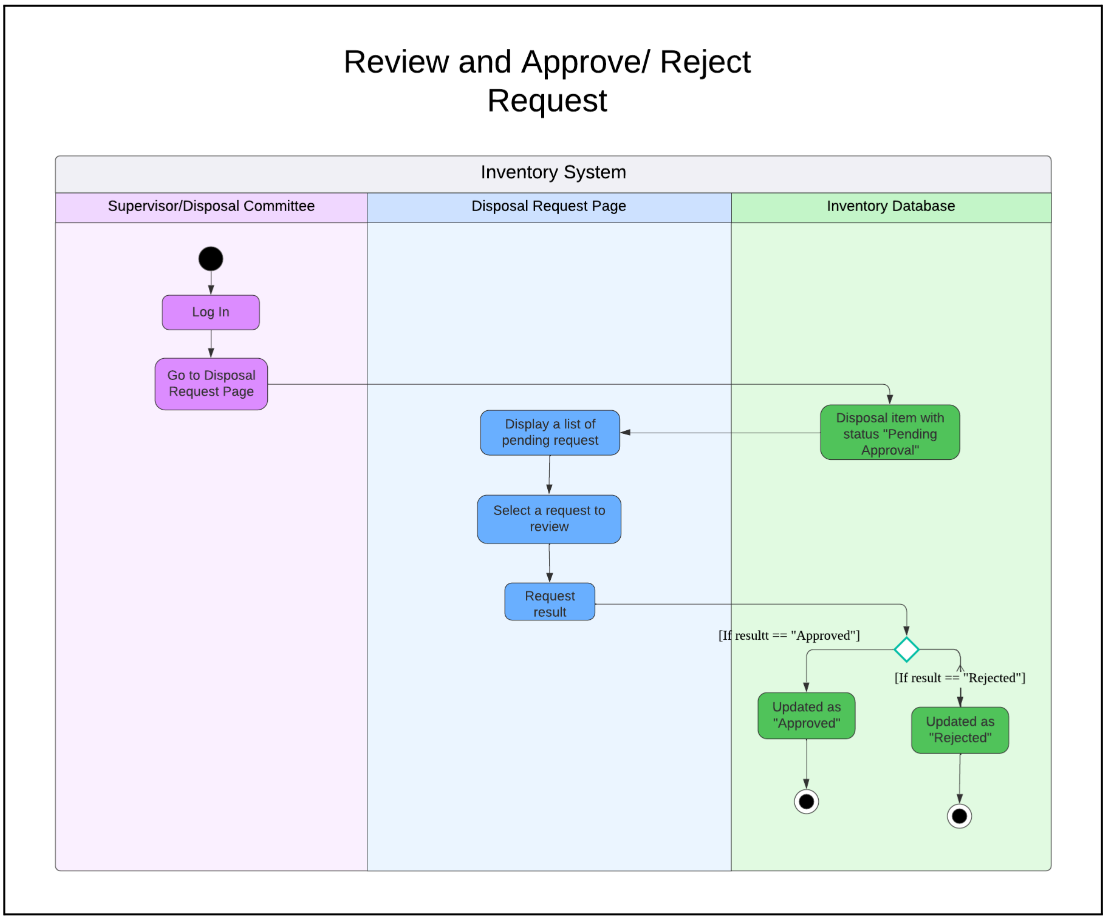


Figure 4.7 Activity Diagram of Review and Approve/Reject Request

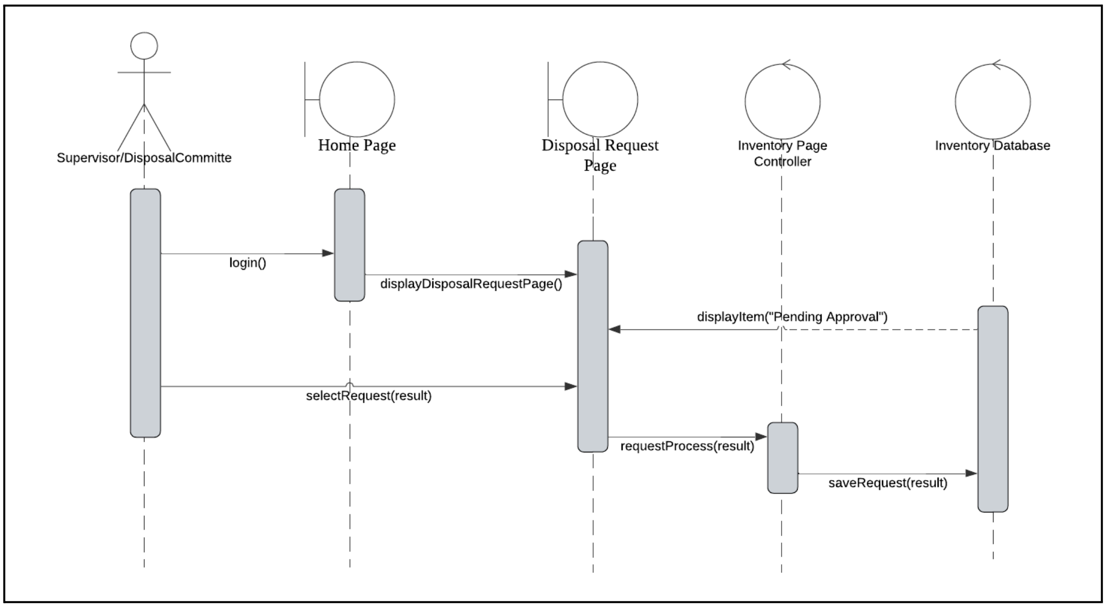


Figure 4.8 Sequence Diagram of Review and Approve/Reject Request

Figure 4.7 and figure 4.8 above shows a workflow that starts with the Supervisor/Disposal Committee navigating to the Disposal Request Page, where the system displays a list of pending requests. The supervisor selects a request to review and either approves or rejects it. Upon approval, the system updates the Inventory Database with the approved disposal item as “Approved” while upon rejection, it marks the request as "Rejected" in the database.

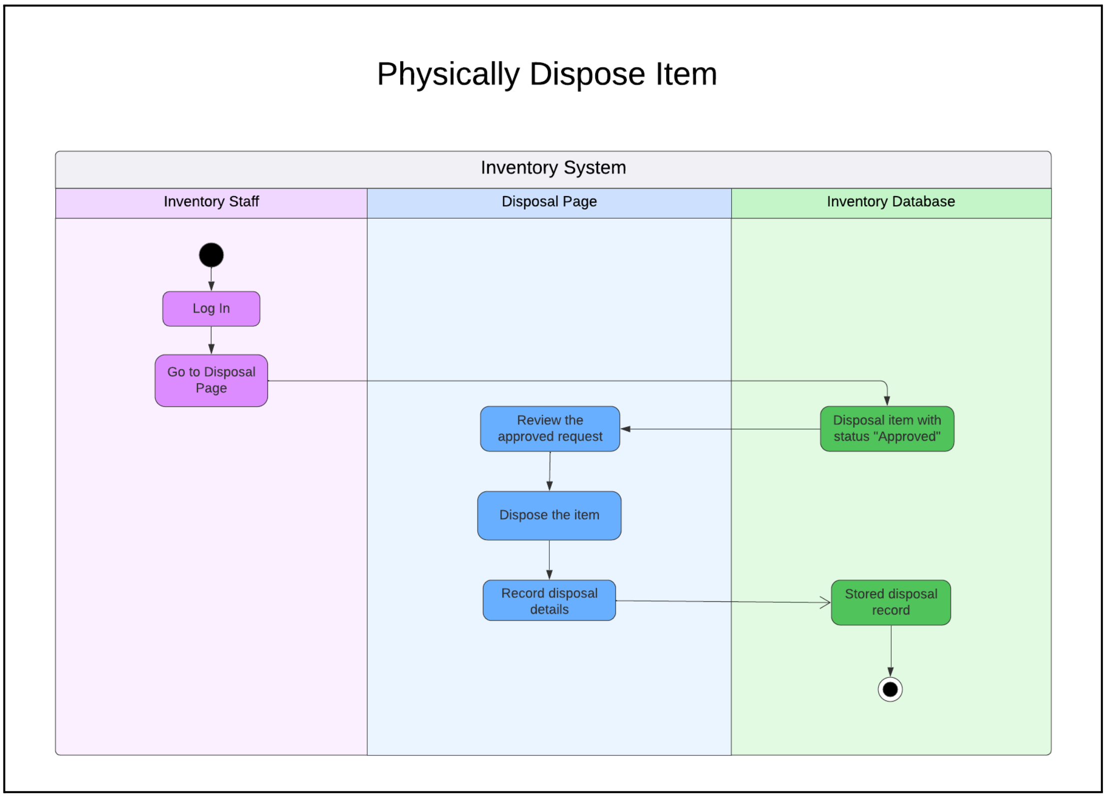


Figure 4.9 Activity Diagram of Physically Dispose Item

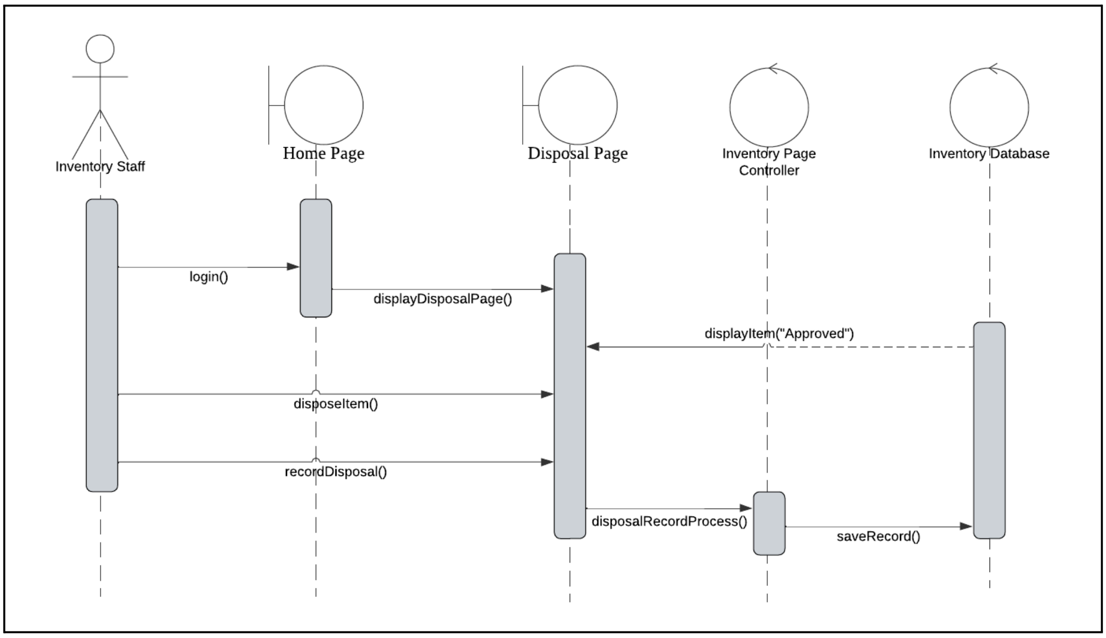


Figure 4.10 Sequence Diagram of Physically Dispose Item

Figure 4.9 and figure 4.10 above shows a workflow that starts with the Inventory Staff logging in and navigating to the Disposal Page to request item disposal. The system then forwards the request for review and approval. Upon supervisor approval, the staff physically disposes of the item, records the disposal details, and the system updates the Inventory Database with the finalized disposal record.

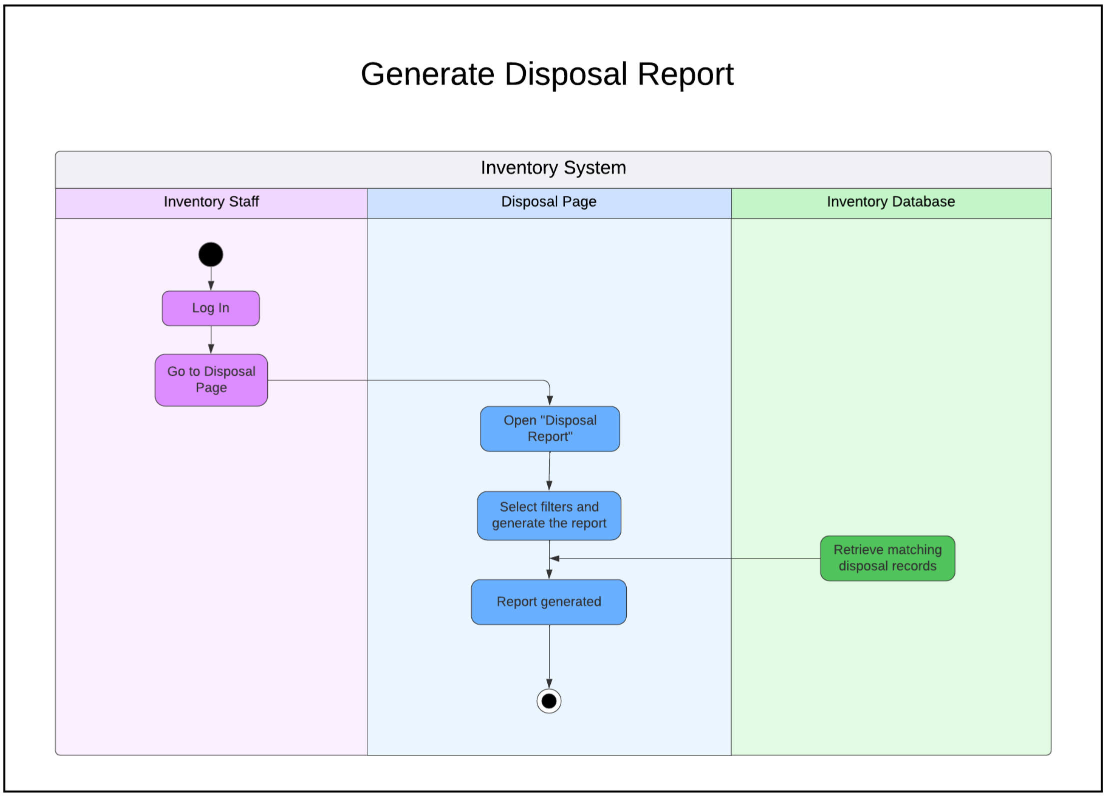


Figure 4.11 Activity Diagram of Generate Disposal Report

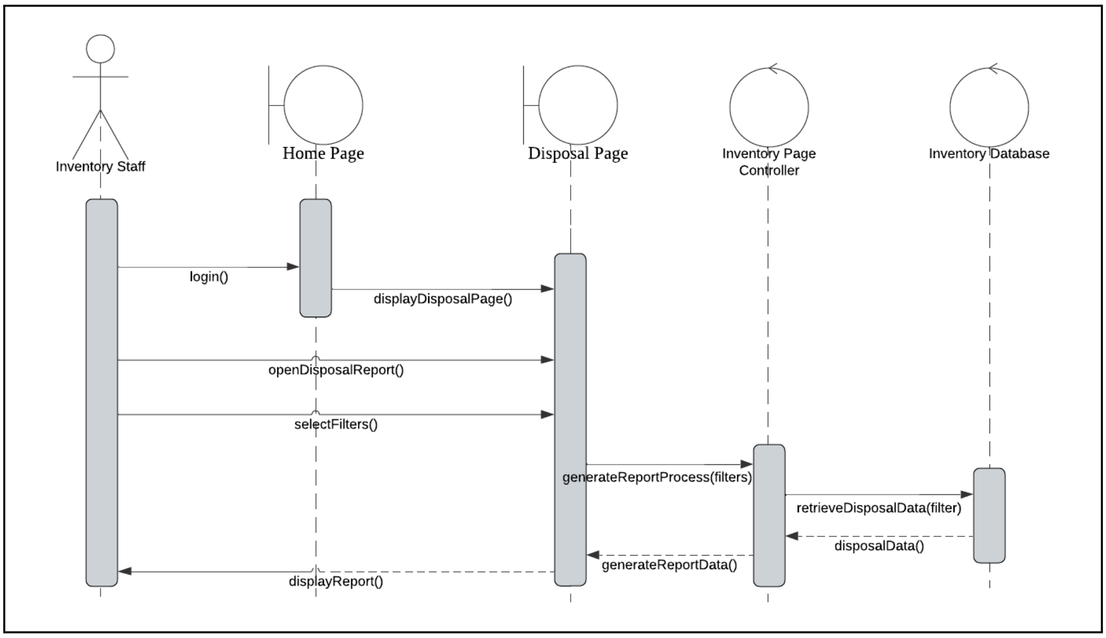


Figure 4.12 Sequence Diagram of Generate Disposal Report

Figure 4.11 and figure 4.12 The process begins when the Inventory Staff opens the "Disposal Report" section and selects specific filters to customize the output. The system then queries the Inventory Database to retrieve matching disposal records based on these filters. Once the relevant data is fetched, the final report is generated and displayed to the user, completing the workflow.

4.7.8 Database Design

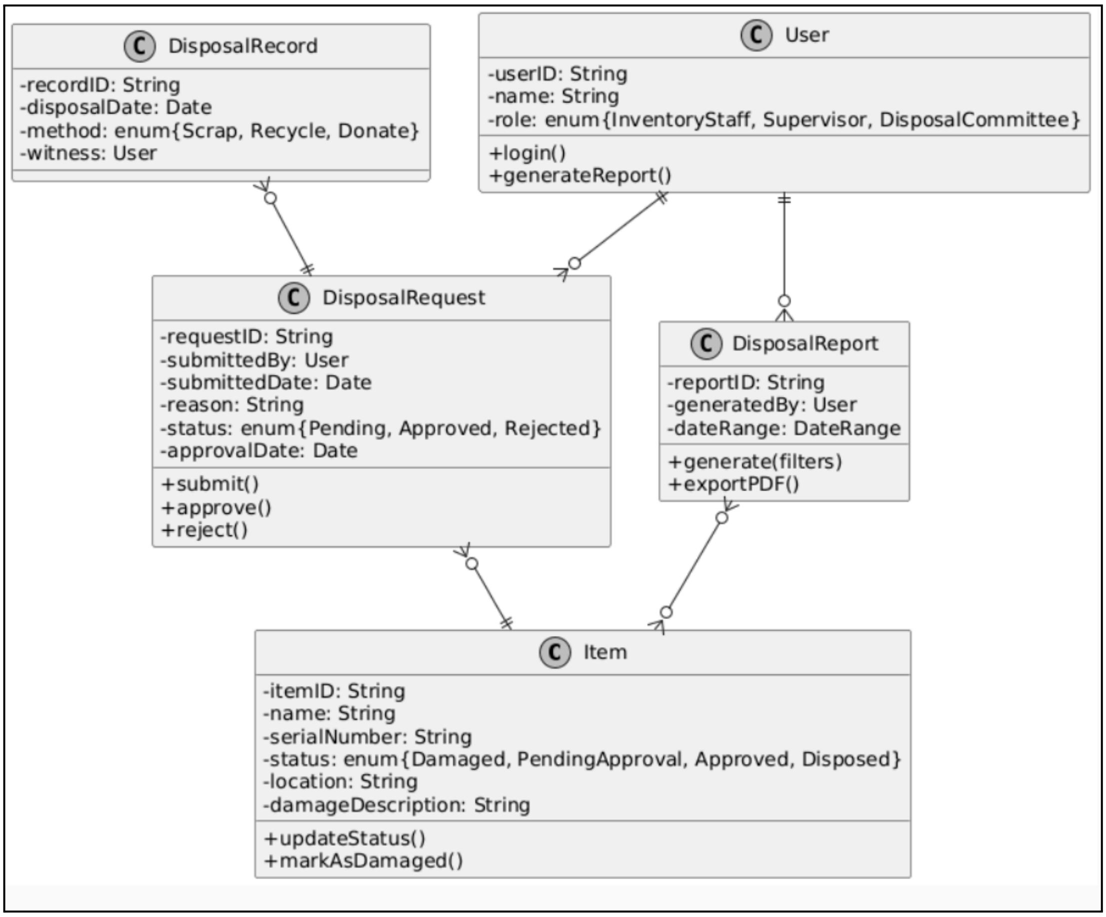


Figure 4.13 Class Diagram of Disposal Inventory System

Figure 4.13 above is the class diagram for this Disposal Inventory System which outlines how the lifecycle of damaged or outdated assets is managed from start to finish with respect to reporting and disposal. The initial step includes capturing the item detail in an Item class. The status of an Item will be tracked based on its condition damaged, under consideration for approval, approved and ultimately disposed of.

The first step after an item has been determined to be damaged is for someone to submit a DisposalRequest to a specific User, such as inventory personnel. A DisposalRequest will contain the reason for the disposal and whether or not approval has been obtained from the appropriate personnel like Supervisor or Disposal Committee.

After the DisposalRequest has received approval, the actual disposal of an item is recorded in the DisposalRecord, which will include the date of disposal, the method of disposal and will name the staff who witnessed the disposal. In addition to recording all disposal activities, staff can also generate and export a DisposalReport that summarizes all disposal activities. The overall working of this diagram indicates a controlled workflow which provides for transparency and authorization of all disposal activities and provides documentation to support the disposal of items.

4.7.9 Interface Design

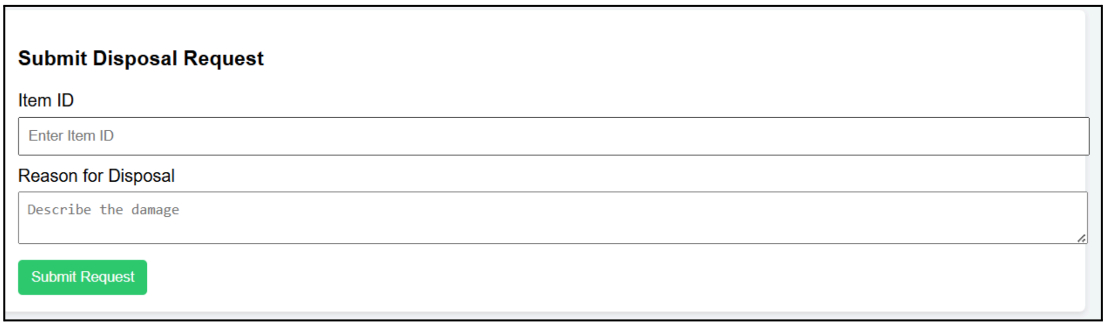


Figure 4.14 Interface of Disposal Request Submission

Figure 4.14 is the input form used by a user to initiate the disposal process. It contains fields for the Item ID and a text box to provide the Reason for Disposal. Once filled, the user clicks the "Submit Request" button to send it for review.

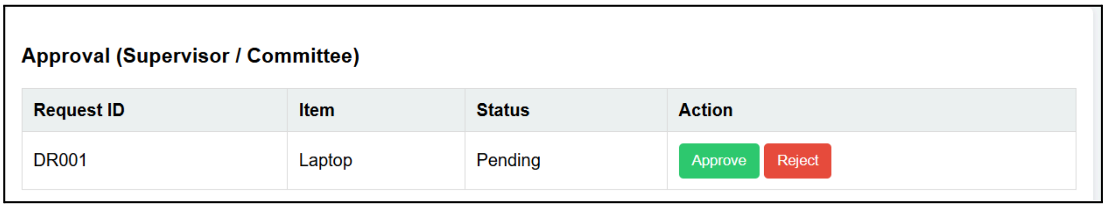


Figure 4.15 Interface of Approval/Rejection from Supervisor or Committee

Figure 4.15 is the management dashboard used by a supervisor or committee member. It displays a table of pending requests that shows the Request ID, the specific item, and the current status. The reviewer can then use the Approve or Reject buttons to take action on the request.

4.8 Chapter Summary

To summarize, this chapter provided a detailed evaluation of UTHM's organizational hierarchy and current "As-Is" architecture. By synthesizing stakeholder requirements gathered through structured interviews, a "To-Be" Enterprise Architecture was proposed to establish a unified, cloud-based ecosystem featuring centralized data management and Single Sign-On (SSO) integration. To bridge the gap between high-level architectural planning and technical implementation, the project has developed specific design artifacts for the UTHM Inventory Subsystem, including Use Case Diagrams and Descriptions to define actor interactions, Activity Diagrams to map procedural workflows, and Sequence Diagrams to illustrate real-time logic. These behavioral models, combined with comprehensive database and interface designs by utilizing SAP BTP Trial Account, The proposed system also ensures that it is structurally sound, user-centric, and fully prepared for the development phase using SAP BTP technology to support the university's digital transformation.

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 Introduction

This chapter focused on the development and implementation of the disposal inventory system prototype, using SAP Build Apps at the primary low code platform which is also known as drag and drop platform to align with UTHM's TO-BE enterprise architecture goals of integration. SAP Build Apps enables rapid development of responsive interfaces with built in connectors to SAP backend services, facilitating smooth data flow for inventory processes such as requesting disposal, supervisor or disposal committee approval and generating reports. The implementation translates from Chapter 4 design including use case diagram, activity diagram, sequence diagram and system architecture that turns into a functional prototype. This chapter will cover how the system is developed, database creation, coding of the main processes and also the chapter summary.

5.2 System Development

Section 5.2 Create Database documents the establishment of the data foundation for the disposal inventory subsystem, utilizing SAP Build Apps' built-in data modeling integrated with SAP HANA Cloud. This individual effort realizes the database design from Chapter 4.8.6.

System Initialization

The SAP Build Apps project was initialized by importing key ERD entities, including the Disposal Request and Approval tables, which were configured as core data models. These models were connected to the SAP HANA Cloud backend to enable real-time synchronization between application interfaces and the centralized database. This setup ensured that disposal activities were supported by a reliable and continuously updated institutional data environment.

Application Development

The interface of this application was developed using the Drag-and-Drop feature of SAP Build Apps. Responsive Layouts were established, allowing for convenient access from multiple devices that are typically used in a university setting like tablets and desktop computers. During this phase of application development, variables were created, and formulas were generated to allow for interactive, dynamic use of the application and scalability to be able to handle the projected increase in the volume of disposal records. Through this design strategy, the application will be able to scale to meet the anticipated increase in volume without negatively impacting performance or ease of use.

System Configuration and Integration

Two connectors were configured within the application and connected to SAP's back-end services. These connectors facilitated the ability to cross-reference disposal requests, stock check records, and repair history data within one centralized data governance area, enabling the validation of accuracy across these three records. Workflow action scripts were developed for lifecycle transition management (request > approved > disposed), as well as automatic updates of the database and notification to the appropriate finance department for the write-off of assets.

5.3 Create Database

Figure 5.1 Database Creation in SAP Build Apps

Figure 5.1 above is a DisposalRequest database for the subsystem was created in SAP Build Apps by defining entities with fields ID, ItemName, Quantity, Reason, Status, and StaffName, directly matching the ERD from Chapter 4 and syncing to SAP HANA Cloud for scalable persistence. This normalized 3NF model supports key functions like request submission and status tracking, with validations ensuring data integrity for disposal workflows, as shown in the Fields preview with "Add New" form integration.

5.4 Coding of the system's main functions/Process

Section 5.4 presents the coding of the disposal inventory subsystem's main processes using SAP Build Apps' visual low-code logic flows, realizing individual use cases like request submission and status approval from Chapter 4. Development focused on three core functions via the Logic Canvas.

Submit Disposal Request

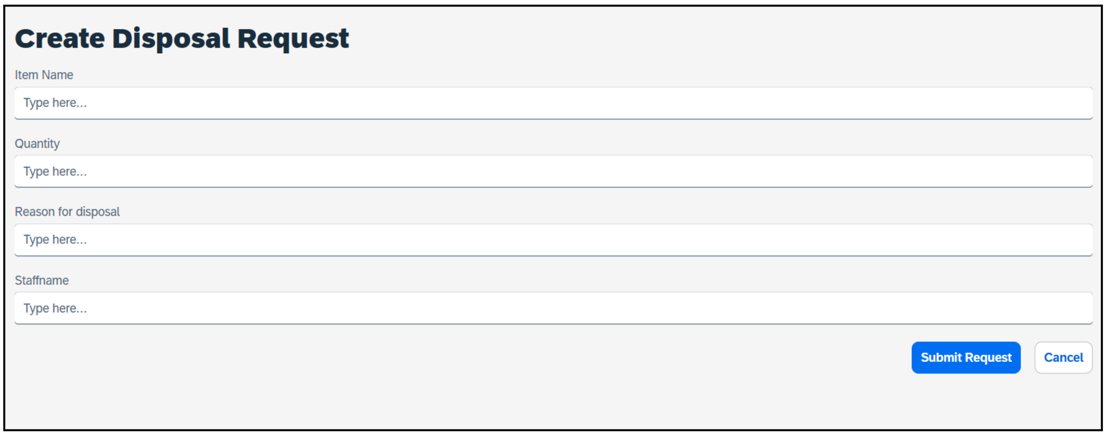


Figure 5.2 Interface of Submit Disposal Request

From figure 5.2 above, Inventory staff have access the Disposal Request page to fill five inputs (ID, ItemName, Quantity, Reason, StaffName), with Status auto-set to "Pending" via logic: CREATE("DisposalRequest", {ItemName: ItemNameInput.text, Quantity: QuantityInput.text, Reason: ReasonInput.text, StaffName: StaffNameInput.text, Status: "Pending"}). Validation ensures IF(QuantityInput.text > 0, Submit, ShowAlert("Invalid quantity")), then refreshes the requests table automatically.

Approve/Reject Request

Figure 5.3 Interface of Submit Disposal Request

Figure 5.3 is the Approval page that filters SELECT(DisposalRequest, Status = "Pending") to display pending requests; tapping Approve executes UPDATE(DisposalRequest, CurrentRow.ID, {Status: "Approved"}), while Reject runs UPDATE(DisposalRequest, CurrentRow.ID, {Status: "Rejected"}). Post-update, the record disappears via table refresh/filter logic, with notifications sent to staff.

5.5 Summary

Through the use of SAP Build Apps, the Prototype of the Disposal Inventory Subsystem in Chapter 5 has completed the process of developing the prototype from Chapter 4's design into a set of working low-code components that align with UTHM's enterprise architecture goals. Major achievements to date include: creating a DisposalRequest data table containing the fields: ID, ItemName, Quantity, Reason, StaffName, and auto-set Status; providing intuitive user interface screens for inventory staff to submit Disposal Requests and Supervisors to approve/disapprove those requests; implementing business logic flows that will allow for end-to-end workflows from the creation of a Disposal Request to updating the Status of the Disposal Request.

CHAPTER 6

CONCLUSION

6.1 Introduction

This final chapter provides a reflective summary of the UTHM Inventory System project. It outlines the primary achievements of the development process, acknowledges the limitations encountered during the implementation in the SAP Build Apps environment, and proposes strategic suggestions for future enhancements. The conclusion reinforces how the system fulfills the original objective of digitizing the manual inventory and its processes at the university.

6.2 System Contribution/Achievement

The UTHM Inventory System has successfully introduced several key improvements to the management of university assets:

- **Centralized Data Management:** The transition from manual records to a structured relational database (On-Device Storage) ensures that user, item, and transaction data are synchronized and easily retrievable.
- **Automation of Approval Workflows:** By implementing sequential logic, the system automatically updates item availability statuses upon clerk approval, reducing administrative workload and preventing data entry errors.
- **Enhanced User Transparency:** Borrowers can now view real-time item availability and report damages directly through the mobile interface, fostering better accountability for university property.
- **Real-time Damage Reporting:** The integration of a dedicated damage reporting module allows the maintenance team to identify and isolate faulty equipment immediately.
- **Better Decision Making and Planning:** The management can make an accurate decision regarding item procurement, replacement strategies, allocation of campus resources and also long term infrastructure planning.

- **Increased Security and Reliability**: Supported by secure network infrastructure, backup systems, and disaster recovery mechanisms, the subsystem ensures that critical inventory data remains protected, available, and recoverable at all times.

6.3 System Constraint

Despite the successful deployment of the prototype, several constraints were identified:

- **Offline Data Dependency:** As the system currently utilizes On-Device Storage, data is localized to the device. Real-time synchronization across multiple different clerk devices would require an external cloud database integration.
- **Cost Implementation:** By implementing SAP Business Technology Platform (BTP) environment requires a significant amount of investment and long-term commitment from the IT steering committee.
- **Technical Skill Requirement:** The system requires personnel with adequate technical knowledge to maintain, configure and troubleshoot the application. Limited expertise in SAP BTP and related technologies may slow down system enhancements and issue resolution.
- **Limited System Integration:** The current prototype operates as a standalone system and has limited integration with existing organizational systems. This may cause data error and reduce efficiency until proper system integration is implemented.

6.4 Future Suggestion

To further evolve the UTHM Inventory System, the following enhancements are recommended:

- **Cloud Backend Integration:** Transitioning the database to a cloud-based service that is also available in SAP BTP to allow real-time data sharing between students and multiple administrative departments.

- **QR Code/Barcode Integration:** Implementing a scanning feature using the device camera to allow users to scan an asset tag, which would automatically pull up the item's details and all the information users need to know.
- **Deep Integration with Financial Systems:** The system should further evolve to integrate directly with the Financial System via API. This would allow the automated calculation of asset depreciation and block the student account in LMS if they fail to return borrowed equipment on time.
- **Internet of Things (IoT) & RFID Tracking:** To automate the stock checking process, future iterations might upgrade from passive QR codes to active RFID tags. By installing the IoT sensor in laboratories, the system could auto detect the position of the high value asset , and no more manual scan of each asset by staff.
- **Predictive Analytics and Smart Dashboard Enhancement:** The system could incorporate predictive analytics powered by SAP analytics tools to forecast asset usage, maintenance cycles, and replacement needs. By analysing historical data from borrowing, repair, and disposal modules, the dashboard can provide insights such as high-demand items, frequently damaged assets, and upcoming procurement requirements. This would support proactive decision-making by administrators and reduce unexpected asset shortages.

6.5 Summary

In summary, the UTHM Inventory System project has successfully demonstrated the efficacy of using low-code development to solve campus administrative challenges. By streamlining the processes and digitizing inventory records, the system provides a robust framework for asset management. While certain technical constraints exist regarding data synchronization, the proposed future suggestions provide a clear roadmap for scaling the application into a comprehensive university-wide solution.

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